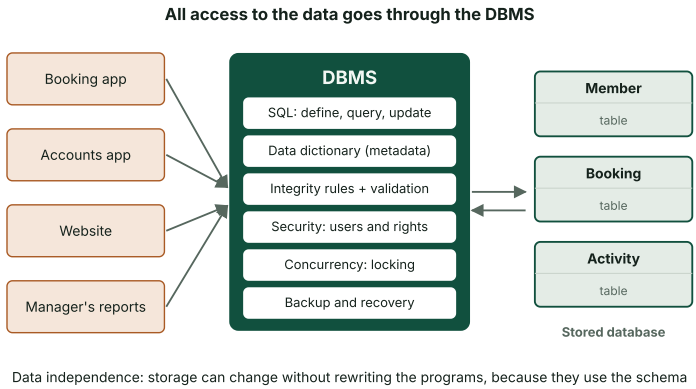
Data management: flat files, relational databases and the DBMS

IS01 FACT FILE

Specification: students should be able to

- Compare flat-file and relational approaches to storing data
- Describe the functions of a database management system (DBMS)
- Explain data independence, data redundancy and data inconsistency

Key words: flat file, relational database, DBMS, data redundancy, data inconsistency, data independence, data integrity, schema, data dictionary, concurrent access, program-data dependence



Application programs reach the data only through the DBMS, which provides SQL, the data dictionary, integrity, security, concurrency control and backup; this gives data independence

Why flat files cause problems

In a flat-file or traditional file-based system the same data is often held in several records or files. This data redundancy wastes storage and, more importantly, leads to data inconsistency: when one copy is updated and another is not, the organisation no longer knows which value is correct. Programs are also tied to the structure of their files (program-data dependence), so a small change to the file can force programs to be rewritten. Flat files remain acceptable for small, single-user lists.

The DBMS

A database management system is software that sits between users or application programs and the stored data. It provides a query language (usually SQL) for defining, updating and retrieving data, keeps a data dictionary describing every table and field, enforces validation and referential integrity, controls access through user accounts and permissions, manages concurrent access with locking and transactions, and supports backup and recovery. Because all access goes through the DBMS, data is controlled centrally and shared safely.

Data independence

Data independence is the separation of the logical view of data used by programs from the way it is physically stored. With physical data independence, storage details such as indexes or disks can change without affecting programs. With logical data independence, new fields or tables can be added and existing applications still work. This reduces maintenance costs and allows the database to evolve as the organisation grows.

T Key terms from the mark schemes

Flat file
A database held as a single table or file with no relationships to other tables.

Data redundancy
The unnecessary storage of the same data item more than once.

Data inconsistency
A situation where copies of the same data item hold different values, so it is unclear which is correct.

Database management system (DBMS)
Software that allows a database to be defined, created, queried, updated and controlled, including security, integrity and concurrent access.

Data independence
The ability to change the way data is stored or structured without having to change the application programs that use it.

Relational database
A database in which data is stored in tables (relations) linked to each other through primary and foreign keys.

Exam focus

'Explain' questions need a point plus its consequence: 'data is stored once (1) so an update only has to be made in one place, avoiding inconsistency (1)'. Do not confuse redundancy (duplication) with inconsistency (disagreeing copies) – examiners treat them as separate points. When asked for DBMS functions, give distinct functions (security, integrity, concurrency, backup, data dictionary, query language) rather than several examples of the same one. Avoid vague words such as 'easier' or 'faster' without saying why.

? Check yourself

1. Define the term data redundancy. [1]
2. Explain how data redundancy can lead to data inconsistency. [2]
3. Describe four functions of a database management system. [4]

Answers: 1. Data redundancy is the unnecessary storing of the same data more than once (e.g. in several records or files) (1). Accept: duplication of data. 2. When the same data is held in several places, an update may be made to one copy but not the others (1) / so the copies no longer agree and the organisation cannot tell which value is correct, e.g. a member's address changed on one booking but not the rest (1). 3. Any four from, with a description: stores, retrieves and updates data, usually via a query language such as SQL (1) / maintains the data dictionary of metadata about tables, fields, types and relationships (1) / enforces integrity: validation rules, entity and referential integrity (1) / provides security through user accounts, passwords and access rights to tables, views or fields (1) / controls concurrent access, e.g. record locking so multi-user updates do not conflict (1) / provides backup and recovery after failure (1) / provides data independence by separating programs from stored data via the schema (1).